UniFind

Campus Lost & Found Website

Business Analysis Requirements Document

Version 1.4 – Approved Requirements Baseline

Field	Details
Project	UniFind – Campus Lost & Found Website
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Role	Business Analyst
Prepared By	M. A. P. S. Perera
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Update	Added FR-26 and US-35 – Claim Found Item; updated scope, actors, business rules, traceability and appendix.
Platform	Web-based website accessed through modern web browsers
Approval Status	Approved by Project Team

Document Change History

Version	Date	Change
1.0	06 October 2026	Original requirements baseline.
1.1	26 October 2026	Added FR-25 Reset Password and US-34 Reset Password. Existing requirement IDs were preserved.
1.3	02 November 2026	Added FR-26 Claim Found Item and US-35; updated project overview, objectives, stakeholders, actors, scope, business rules, backlog summary, traceability, assumptions and Appendix B. Version prepared for review.
1.4	03 November 2026	Approved updated requirements baseline, including FR-26 and US-35 – Claim Found Item.

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1. Project Overview

The UniFind – Campus Lost & Found Website is a web-based platform designed to help university students report lost and found items, search for relevant reports, upload item images, identify possible matches using image matching technology, and communicate with other users regarding potentially recovered items. The website also provides administrative functions for managing users, reports, flags, and disputes. Registered users can reset a forgotten password through a verified password-reset process. Registered users can also submit a claim for a found item they believe belongs to them, subject to the agreed claim-review process.

2. Problem Statement

Students often face difficulties when trying to recover lost belongings within the university environment. Traditional methods such as notices, social media posts, and word-of-mouth can make it difficult to identify matching items and communicate with the correct owner. The proposed website provides a centralized digital platform where students can report, search, match, and communicate about lost and found items. Users who forget their passwords also need a secure way to regain access to their accounts. A structured claim-submission process is also needed so that a user can request an item they believe belongs to them.

3. Project Objectives

- Allow students to report lost items.
- Allow students to report found items.
- Provide an easy way to search for lost and found items.
- Allow users to upload item images.
- Identify possible matching items using image matching.
- Allow users to communicate about possible matches.
- Allow users to submit claims for found items they believe belong to them.
- Allow users to flag inappropriate content or behavior.
- Allow administrators to manage users, reports, flags, and disputes.
- Provide a reliable, responsive, and user-friendly campus lost-and-found website.
- Allow registered users to securely reset forgotten passwords after identity verification.

4. Stakeholders

Stakeholder	Role / Interest
Students	Main users who report, search, match, claim, and recover lost or found items.
System Administrator	Manages users, reports, flags, disputes, and reviews claims according to the agreed workflow.
Project Team	Analyzes, designs, develops, tests, and maintains the website.
University	Provides the campus environment, policies, and governance context.
Business Analyst	Elicits, analyzes, documents, prioritizes, and validates requirements.
QA / Tester	Verifies that implemented website functionality satisfies acceptance criteria.

5. System Actors

Actor	Responsibilities
Student / User	Register, log in, manage profile, reset a forgotten password, report items, upload images, search, view matches, submit claims for found items, communicate, flag content, submit disputes, and provide feedback.
Administrator	Manage users and reports, review claims, flags and disputes, and take appropriate administrative action.
Image Matching System	Analyze and compare item images and return possible matches with similarity information.

Notification System	Deliver relevant notifications such as possible matches, messages, and status updates.
Web Browser	Provides the interface through which users access the web-based website.

6. Project Scope

6.1 In Scope

- User registration and authentication
- User profile management
- Password reset for registered users
- Lost item reporting
- Found item reporting
- Item image upload and validation
- Item search and filtering
- Item details and status management
- Image-based matching
- Possible match results
- Claim submission for found items
- Review of submitted claims by an authorized person, subject to the agreed workflow
- Communication between authorized users
- Notifications
- Flagging users or reports
- Dispute submission and resolution
- Reviews / feedback
- Administrative management

6.2 Out of Scope

- Physical delivery of recovered items
- Buying or selling items
- Payment processing
- External marketplace integration
- Advanced facial recognition
- Automatic physical verification of ownership

6.3 Platform and Access

The system will be developed as a web-based website. Users will access the website through a modern web browser using an internet connection. The website will provide responsive and user-friendly interfaces suitable for desktop and mobile screen sizes.

7. Functional Requirements

ID	Functional Requirement	Priority
FR-01	The system shall allow students to create a UniFind account by providing their	Must

	name, university email address, password, and password confirmation. The system shall validate the entered information and prevent duplicate registrations.	
FR-02	The system shall allow registered users to log in and log out.	Must
FR-03	The system shall allow users to manage their profiles.	Should
FR-04	The system shall allow users to report a lost item.	Must
FR-05	The system shall allow users to report a found item.	Must
FR-06	The system shall allow users to upload images of lost/found items.	Must
FR-07	The system shall allow users to edit and delete their own item reports.	Must
FR-08	The system shall display lost and found item listings.	Must
FR-09	The system shall allow users to search for items by keywords.	Must
FR-10	The system shall allow users to filter items by category, location and date.	Should
FR-11	The system shall compare uploaded item images with existing item images.	Must
FR-12	The system shall display possible image matches ranked by similarity.	Must
FR-13	The system shall allow users to view detailed information about an item.	Must
FR-14	The system shall allow users to contact another user regarding a possible match.	Should
FR-15	The system shall allow users to send and receive messages.	Should
FR-16	The system shall notify users about relevant updates or possible matches.	Should
FR-17	The system shall allow users to flag inappropriate reports or users.	Should
FR-18	The system shall allow users to submit disputes.	Should
FR-19	The system shall allow users to provide reviews/feedback after an interaction.	Could
FR-20	The system shall allow administrators to manage users.	Must
FR-21	The system shall allow administrators to review flagged content/accounts.	Must
FR-22	The system shall allow administrators to review and resolve disputes.	Must
FR-23	The system shall allow administrators to warn, suspend or deactivate accounts.	Must
FR-24	The system shall maintain item statuses such as Lost, Found, Matched, Returned and Closed.	Must
FR-25	The system shall allow registered users who have forgotten their passwords to reset them after completing an approved identity-verification process. The system shall validate the new password and securely update the stored password.	Must
FR-26	The system shall allow an authenticated user to submit a claim for a found item they believe belongs to them. The system shall validate the required claim information, associate the claim with the submitting user and the selected found-item report, and store the claim for review by an authorized person. The claim's initial status and review workflow shall follow the process agreed by the project team.	Must

8. Non-Functional Requirements

ID	Non-Functional Requirement	Priority
NFR-01	Normal user requests should receive a response within approximately 2–3 seconds under normal conditions.	Should
NFR-02	Image matching should provide results within an acceptable time.	Should
NFR-03	User passwords must be securely hashed and never stored as plain text.	Must
NFR-	Protected features must require authentication.	Must

04		
NFR-05	Users must only be able to modify their own reports.	Must
NFR-06	Private user information and conversations must only be accessible to authorized users.	Must
NFR-07	The system should minimize unexpected failures and data loss.	Must
NFR-08	The website interface should be simple, responsive, consistent, and easy for students to navigate on desktop and mobile devices.	Must
NFR-09	The website should work correctly on supported modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.	Should
NFR-10	The system should maintain accurate and consistent data.	Must
NFR-11	Important administrative actions should be recorded.	Should
NFR-12	Uploaded images should be validated for supported format and size.	Must
NFR-13	The system should display clear error messages when invalid information is entered.	Must
NFR-14	The system should be maintainable and structured for future development.	Should

The password-reset feature must comply with NFR-03, NFR-04, NFR-06, and NFR-13 where applicable. Claim submission must comply with NFR-04, NFR-06, NFR-10, and NFR-13 where applicable.

9. Business Rules

Rule ID	Business Rule
BR-01	Only students who meet the agreed university email requirements may register for a UniFind account, and only registered users can submit lost or found item reports.
BR-02	A user can edit or delete only their own reports.
BR-03	An item report must contain all required information before submission.
BR-04	An item may have a status of Lost, Found, Possible Match, Matched, Returned, or Closed.
BR-05	A user must not be able to access another user's private conversations.
BR-06	Administrators can review reports that have been flagged.
BR-07	Administrators can take action against accounts that violate system rules.
BR-08	A password may be reset only after the user completes the approved identity-verification process.
BR-09	Only authenticated users may submit a claim. Each claim must refer to a valid found-item report and be associated with the submitting user. Only an authorized person may review or update a claim. The initial claim status and claim-decision workflow must be agreed by the project team.

10. Flagging Rules

A user or item report may be flagged when:

- A user submits false information.
- A user repeatedly creates misleading reports.
- A user uploads inappropriate content.
- A user harasses another user.
- A user behaves abusively.
- A user attempts fraudulent activity.
- A user violates applicable university rules.

Flagging Process:

1. User submits flag
2. System records flag
3. Administrator reviews flag
4. Administrator investigates
5. Administrator makes a decision
6. Action is taken if required
7. Flag is resolved

11. Administrative Actions

- Review flagged users.
- Review flagged reports.
- Review submitted item claims according to the agreed claim workflow.
- Review disputes.
- Send warnings.
- Temporarily suspend accounts.
- Permanently deactivate accounts.
- Remove inappropriate reports.
- Resolve disputes.

12. Dispute Management

A user may submit a dispute when they believe an ownership claim is incorrect, an item was not returned as expected, another user provided false information, there was inappropriate behavior, or there is disagreement regarding a reported item.

Status	Meaning
Open	Dispute has been submitted and is awaiting review.
Under Review	Administrator is investigating the dispute and evidence.
Resolved	Administrator has completed the review and recorded a decision.

Possible outcomes include No Action, Warning, Account Suspension, Account Deactivation, and Report Removed.

13. User Story Backlog Summary

ID	Epic	User Story	Priority
US-01	User Management	Register Account	Must
US-02	User Management	Login	Must
US-03	User Management	Logout	Must
US-04	User Management	Manage Profile	Should
US-05	Lost & Found Item Management	Report Lost Item	Must
US-06	Lost & Found Item Management	Report Found Item	Must
US-07	Lost & Found Item Management	Upload Item Image	Must
US-08	Lost & Found Item Management	View Item Details	Must
US-09	Lost & Found Item Management	Edit Item Report	Must
US-10	Lost & Found Item Management	Delete Item Report	Must
US-11	Lost & Found Item Management	Update Item Status	Must
US-12	Image Matching	Upload Image for Matching	Must
US-13	Image Matching	Compare Item Images	Must
US-14	Image Matching	Display Possible Matches	Must
US-15	Image Matching	View Match Details	Must
US-16	Search & Discovery	Search Items	Must
US-17	Search & Discovery	Filter Items	Should
US-18	Search & Discovery	View Search Results	Must
US-19	Communication & Notifications	Contact User	Should
US-20	Communication & Notifications	Send Message	Should
US-21	Communication & Notifications	Receive Message	Should
US-22	Communication & Notifications	Receive Notifications	Should
US-23	Reviews & Feedback	Submit Feedback	Could
US-24	Reviews & Feedback	View Feedback	Could
US-25	Administration & Disputes	Manage Users	Must
US-26	Administration & Disputes	Review Item Reports	Must
US-27	Administration & Disputes	Review Flagged Accounts	Must
US-28	Administration & Disputes	Flag User or Report	Should
US-29	Administration & Disputes	Submit Dispute	Should
US-30	Administration & Disputes	Review Dispute	Must
US-31	Administration & Disputes	Resolve Dispute	Must
US-32	Administration & Disputes	Suspend User	Must
US-33	Administration & Disputes	Deactivate User	Must
US-34	User Management	Reset Password	Must
US-35	Lost & Found Item Management	Claim Found Item	Must

14. Sprint 1 Candidate Stories

The following stories are recommended as Sprint 1 candidates. Final Sprint 1 scope should be agreed with the development team and project stakeholders.

- Register Account
- Login
- Report Lost Item
- Report Found Item
- Upload Item Image
- View Item Listings
- Search Items
- View Item Details
- Basic Image Matching

15. Requirement Traceability

Requirement(s)	Epic	Representative Stories	Business Area
FR-01 / FR-02	User Management	Register Account / Login	Account access
FR-03	User Management	Manage Profile	Profile management
FR-04 / FR-05	Lost & Found Item Management	Report Lost Item / Report Found Item	Item reporting
FR-06 / FR-11 / FR-12	Image Matching	Upload / Compare / Display Matches	Image matching
FR-08 / FR-09 / FR-10	Search & Discovery	View Listings / Search Items / Filter Items	Discovery
FR-13	Lost & Found Item Management	View Item Details	Item details
FR-14 / FR-15 / FR-16	Communication & Notifications	Contact / Message / Notifications	Communication
FR-17 / FR-18	Administration & Disputes	Flag / Submit Dispute	Governance
FR-20–FR-24	Administration & Disputes	Manage / Review / Resolve	Administration
FR-25	User Management	US-34 Reset Password	Account recovery
FR-26	Lost & Found Item Management	US-35 Claim Found Item	Claim submission and review

16. Assumptions and Constraints

16.1 Assumptions

- The primary users are university students.
- Users have access to a modern web browser and internet connection.
- Users provide reasonably accurate item information.

- The image matching functionality can be implemented using an appropriate technical approach.
- Administrative users are authorized by the university/project context.
- Password reset will use an identity-verification method agreed by the project team.
- Claim submission will be available to authenticated users and will follow a claim-review process agreed by the project team.

16.2 Constraints

- Limited project time and resources.
- Image quality and the selected image-matching approach may affect matching accuracy.
- University privacy, security, and usage policies must be considered.
- The project focuses on core campus lost-and-found functionality rather than a general marketplace.
- The website must remain usable across supported modern browsers and common screen sizes.
- The specific password-reset verification method must be confirmed before implementation.
- The claim status values, evidence/information required from a claimant, and claim review/decision workflow must be confirmed by the project team before final implementation.

Requirements should be reviewed and agreed before development. Changes after approval should be documented, assessed for impact, and communicated to the project team.

17. Requirements Approval

Version 1.4, including the requirements additions and updates introduced in Versions 1.1 and 1.3, has been reviewed and approved by the project team on 03 November 2026. The approval record is provided below.

Role	Responsibility	Name	Approval / Status	Date
Project Manager	Reviews and approves project scope and requirements	P. D. M. Wanigasekara	Approved – Version 1.4	03 November 2026
Business Analyst	Prepares, documents, and maintains project requirements	M. A. P. S. Perera	Prepared / maintained	02 November 2026
QA Engineer	Reviews requirements for clarity, completeness, and testability	K. M. D. Tishani	Approved – Version 1.4	03 November 2026
Developer 1	Reviews requirements for technical feasibility	W. D. Sampath	Approved – Version 1.4	03 November 2026
Developer 2	Reviews requirements for technical feasibility	H. D. M. Rukshani	Approved – Version 1.4	03 November 2026
Developer 3	Reviews requirements for technical feasibility	W. R. A. P. Sakura	Approved – Version 1.4	03 November 2026

APPROVAL STATUS: Version 1.4 is the approved requirements baseline for the UniFind – Campus Lost & Found Website. The project team approval date is 03 November 2026.

Appendix A. US-34 Reset Password

Requirement ID: US-34 (existing user story ID; US-19 remains Contact User).

User Story: As a registered student, I want to reset my password if I forget it so that I can regain access to my account.

Related Functional Requirement: FR-25

Related Non-Functional Requirements: NFR-03, NFR-04, NFR-06, NFR-13

Acceptance Criteria

- The user can access the Reset Password option from the login page.
- The user can provide the required account identification information.
- The system verifies the user's identity before allowing a password reset.
- After successful verification, the user can enter and confirm a new password.
- The system validates the new password according to the approved password requirements.
- The new password is securely stored, and the previous password is replaced.
- The system displays an appropriate success message after a successful reset.
- The system displays appropriate error messages for invalid account details, failed verification, or invalid passwords.
- Unauthorized users cannot reset another user's password.

Implementation Notes

- Use a secure identity-verification process, such as a time-limited reset link or code if email-based recovery is approved.
- Do not allow a password reset based only on knowing a username or email address.
- Do not store passwords in plain text.
- Record the chosen verification method and any agreed password policy in implementation details.

Appendix B. US-35 Claim Found Item

Requirement ID: US-35 (new user story ID; US-20 remains Send Message).

Related Functional Requirement: FR-26

Related Non-Functional Requirements: NFR-04, NFR-06, NFR-10, NFR-13

User Story: As a registered student, I want to submit a claim for a found item that I believe belongs to me so that I can request its return.

Acceptance Criteria

- An authenticated user can submit a claim for a selected found-item report.
- The user must provide all claim information required by the project team.
- The system validates the submitted claim information before saving it.
- The system verifies that the selected found-item report exists and is eligible to receive a claim under the agreed rules.
- The claim is associated with the correct submitting user and found-item report.
- The system saves the claim details and makes them available to an authorized person for review.

- The claim is assigned an initial status defined by the team-approved claim workflow.
- The system displays appropriate success and error messages.
- Unauthenticated users cannot submit claims, and users cannot access another user's private claim information.

Implementation and Clarification Notes

- The existing Claim Item frontend was reported as completed in a previous sprint; the remaining planned work is backend implementation and integration with that existing interface.
- The team must confirm the information/evidence a claimant is required to provide.
- The team must confirm the initial claim status and the claim review, approval/rejection, and notification workflow.
- The system must enforce authentication and protect claim information from unauthorized access.
- Do not assume a particular claim status or approval process until it is agreed and documented.